Theodor Vararu

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Software engineer with a 13-year track record of shipping national infrastructure at NHS, DfE, MoJ, GDS. Author of alphagov/accessible-autocomplete (960 stars, 620k downloads per month). Credited for defining "Accessibility acceptance criteria" at the Government Digital Service.

Experience

Shopify *Senior Engineer* *Sep 2026 – Present*

- **Agentic commerce**: Building UCP and agentic checkout so AI agents can buy from millions of Shopify merchants.

HM Revenue & Customs *Applied AI Architect (Contract)* *Jul 2026 – Sep 2026*

- **RIO & AI**: Advised a major transformation programme on embedding AI into HMRC's tax compliance operations: reviewing delivery plans, accelerating a pilot, and shaping the technical approach for intelligence-led risk identification.

National Health Service *Tech Lead (Contract)* *Jan 2023 – Apr 2026*

- **Manage vaccinations in schools**: Mavis is the national service replacing paper consent forms and recording school-age vaccinations (HPV, flu, MenACWY, Td/IPV 3-in-1). I was the hands-on tech lead from the early Ruby on Rails monolith through to public launch and rollout to 9 immunisation teams, 5,700+ schools, and 1M+ patient records. Built the NHS integrations (PDS, NHS Imms API, CIS2 SSO), observability stack (Splunk via custom Lambda), patient-facing consent journey, and offline-first PWA, across 1,700+ commits as the #2 contributor (I personally sourced the engineer who would take #1 from me). By April 2026, on track to be the most widely used system of its kind in England.
- **Mavis reporting**: Architected and built a Python Flask microservice to expose tabular data and CSV exports for NHS commissioners, with emphasis on materialised views, caching, queues, and PostgreSQL performance optimisation.
- **Gromit**: Built a multi-tenant RAG Slack agent that searches Mavis code and team chat history with source permalinks. Initially in Python and LlamaIndex, ported to Bun single-binary with native ONNX embeddings via bun:ffi, Postgres pgvector + ParadeDB hybrid search, and Claude Agent SDK. The lovable dog is now a core member of the team, helping to draft 19+ Mavis release notes, onboarding new team members, and being spun out as a SaaS product.

Department for Education *Tech Lead (Contract)* *Jan 2022 – Dec 2022*

- **Find a Lost TRN**: Self-service for 500k teachers in England to recover their TRN, the unique number used to apply for jobs or claim pension. Led 2-6 engineers from hello world to public beta and built the OIDC SSO client now part of Get an Identity. Championed and built the public /performance dashboard. The service processes 50k recoveries per year, with the median user getting their TRN in 38 seconds (before: many days).
- **Teaching Regulation Authority**: Bootstrapped two national services: Apply for Qualified Teacher Status, and Refer Serious Misconduct. Wrote ADRs, interviewed developers, scaffolded the apps, deploy pipeline, CI, and path to live.

Ministry of Justice *Senior Developer (Contract)* *Sep 2021 – Dec 2021*

- **Manage Supervisions (HMPPS)**: Brought in for a 2-week accessibility audit and fixes. A few days in, after witnessing my throughput, the delivery manager offered a 3-month retainer to help them deliver on their roadmap.

Meta *Software Engineer* *Jan 2021 – Jul 2021*

- **Workplace**: Accessibility squad on Meta's enterprise comms product (7M+ paid users incl. Spotify, Starbucks, Nestlé). Triaged and fixed issues toward full WCAG 2.2 compliance; wrote the team's accessibility statement.
- **Empathy Lab**: Designed and built prototypes for an accessibility testing tool, a browser extension that simulates disabilities and common assistive technology interactions; pitched as "VoiceOver for software engineers."

Department for Education *Tech Lead / Senior Developer (Contract)* *Sep 2018 – Nov 2020*

- **Apply for Teacher Training**: Led development of the candidate-facing side of the national service handling the UK's entire teacher recruitment pipeline, from hello world to public beta in Nov 2019 (Ruby on Rails, Azure). The 2020 cycle handled over 100k applications. Mentored a team of 5 engineers.
- **Find Postgraduate Teacher Training**: Built features on the public catalogue used to search across 10,000+ courses from 230+ training providers in England (C#, Ruby on Rails). Modernised the DevOps pipeline with Docker, Terraform, and Cypress E2E tests; prototyped Node.js + GraphQL alternatives presented to the architecture group.

Gamesys *Senior Developer (Contract)* *Mar 2018 – Sep 2018*

- **Bundle performance**: Cut bundle size by 33% on a React/Node SPA serving 260k MAU by removing moment.js and trimming lodash. Wrote performance documentation to guide future work.

Graphit Systems *Senior Developer (Contract)* *Aug 2017 – Feb 2018*

- **Online shop with 3D configurator**: Sole frontend engineer on an inherited React.js + Three.js codebase with no handover. Got up to speed fast and shipped a redesign, Stripe payments, and legacy browser fixes.

Government Digital Service *Senior Developer (SC Security Clearance)* *Mar 2016 – Aug 2017*

- **Accessible autocomplete**: Built the most widely used open-source combobox component for GOV.UK services. Iterated based on feedback from research with users with disabilities, and an audit by the Digital Accessibility Centre. Gave a talk at Accessibility London. Credited in the GDS blog for defining "Accessibility acceptance criteria".
- **Government as a Platform**: Worked across the programme on multiple products: a serverless JSON API to generate country data based on the OpenRegister, accessible webchat, data visualisations using D3.js. Managed 2 engineers.
- **GOV.UK**: Rebuilt formats as part of the Migration programme, moving from the legacy publisher to a microservices pipeline. Provided oncall support during second-line shifts, and managed deployments of national infrastructure.

Timecounts *Senior Developer* *May 2015 – Feb 2016*

Workangel *Lead Developer* *Sep 2014 – Apr 2015*

ReadForward *Lead Developer* *Mar 2013 – Sep 2014*

Education

- **London South Bank University** London, UK
  Bachelor of Engineering, Computer Systems and Networks. Transferred from University of Bucharest *2014 – 2016*

Recent Projects

- **keepthingsopen.com (2026)**: I co-wrote an open letter to NHS leadership, asking them to keep their code open. It received over 2,200 signatures from GDS founders (James Stewart, Dafydd Vaughan), notable public figures (Matt Hancock, Cory Doctorow), and many doctors and technologists in government. DSIT released a statement taking the same position, and the letter has been covered by The Register, Health Service Journal, and New Scientist.
- **hasteOS (2025 – 2026)**: I'm building a native macOS Swift menu bar app that uses a local LLM to anticipate the user's next action. It interrogates the accessibility tree, generates a prompt with what's visible for a local and private LLM, and then highlights an element with a prominent border. Tab key accepts the suggestion.
- **braindance (2026)**: I'm building a spatial computing shell for VITURE XR glasses. It captures desktop screens via a zero-copy GPU pipeline and renders them as head-tracked 3D windows. Multi-threaded Rust + Bevy ECS, FFI bindings to the VITURE SDK, and a bespoke 3DoF-to-6DoF tracking pipeline (OpenCV ChArUco + PnP, IMU fusion).
- **tuicraft (2026)**: I built a WoW chat client for the terminal, designed for OpenClaw and other AI agents to play the MMO. Built with Bun, 100% tested, while heavily leveraging AI code generation. Open source, AGPLv3.
- **zp-voicemod (2026)**: I reverse engineered the ZERO PARADES game and wrote a Python pipeline that trains a voice model (hubert-bsqvae-small) on a target voice & patches every dialogue line with your voice. Open source, GPLv3.
- **untitled.ink (2025 – 2026)**: I'm building an interactive fiction platform, featuring a Claude Code-style agentic editor, the Ink markup language, hot-reloading, and video/image generation & editing on Replicate or ComfyUI.

Skills

- **Languages**: TypeScript, JavaScript, Ruby, Python, Rust, Swift, C#
- **Tools**: Bun, React.js, Next.js, Node.js, Astro.js, Flask, Ruby on Rails, Hotwire, Tailwind, PostgreSQL, Redis, SQLite
- **Infrastructure**: AWS, Azure, Docker, Terraform, GitHub Actions, Kamal, DigitalOcean, Vultr, CI, CD
- **Expertise**: Accessibility (WCAG), testing (Playwright, RSpec, Jest, TDD), security, performance, design systems
- **AI / ML**: RAG, ComfyUI, LlamaIndex, Ollama, embeddings, Claude, Codex, MCP, pgvector, OpenClaw, OpenCV